Dr. RVR NRI INSTITUTE OF TECHNOLOGY

Python ABL - COMPLETE CONSOLIDATED REPORT

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Roll No: 26000985

Branch & Section: CSM-2

GitHub: Python-Activity-Based-Learning

Circular Date: 14-10-2026

This Report Contains for Each Case (1-10):

1. Problem Statement
2. Objectives
3. Assumptions
4. Algorithm (Detailed)
5. Flowchart (Visual Diagram)
6. Pseudocode
7. Source Code Ref
8. Test Cases
9. Output Screenshots
10. Conclusion

Evaluation: 10 Marks - All requirements as per Circular Points 2,5,6

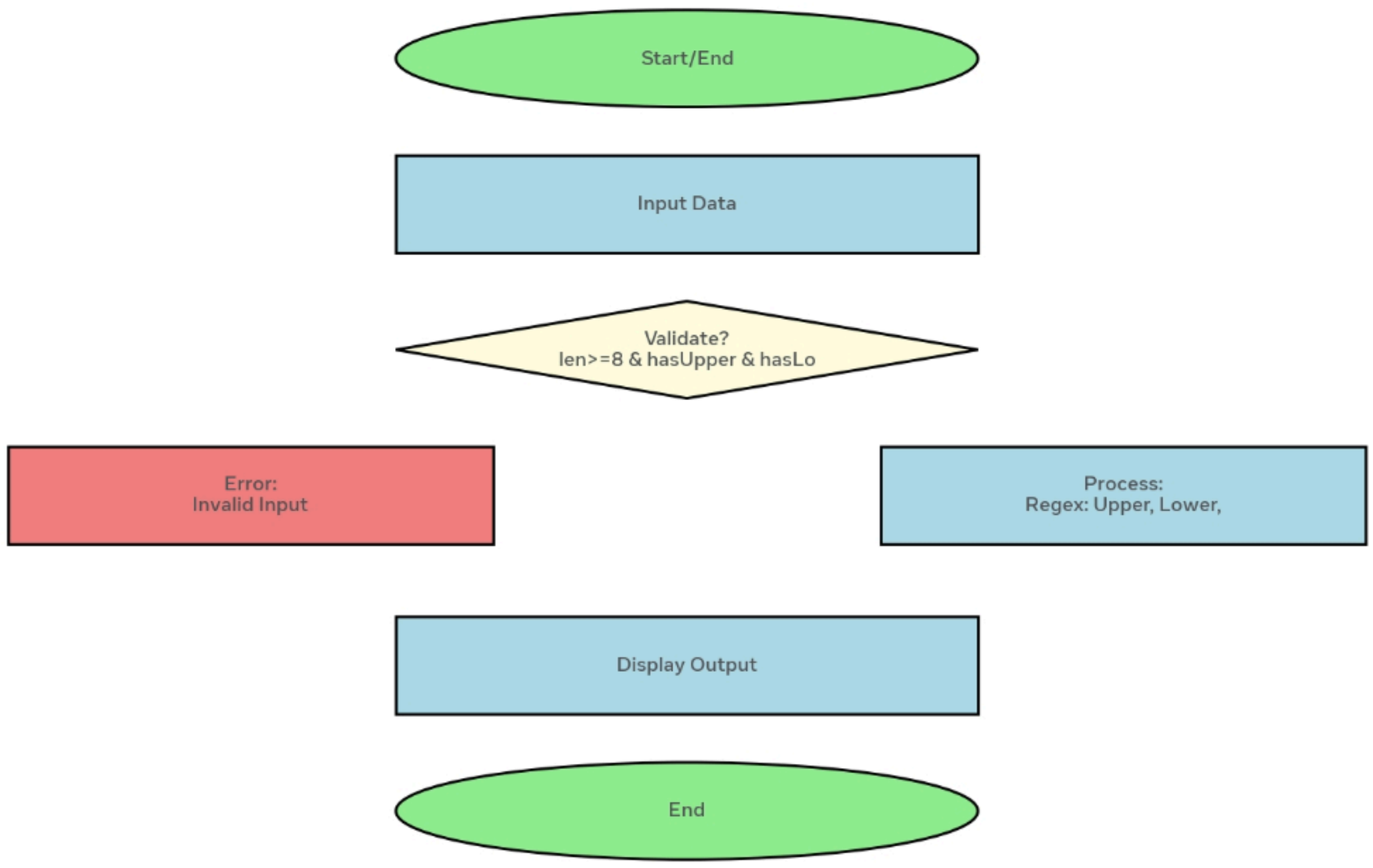
Case 1: Password Validator - 26000985 - CSM2

Problem: Password Policy Validation

ALGORITHM:

1. Start
2. Input data as per Regex: Upper, Lower, Digit, Special
3. Validate: len>=8 & hasUpper & hasLower & hasDigit & hasSpecial
4. Process using Regex: Upper, Lower, Digit, Special
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_1():
    READ input_data
    IF validate(input_data) THEN
        result = process(Regex: Upper, Lower, Digit, Special)
        PRINT result
    ELSE
        PRINT 'Invalid - len>=8 & hasUpper & hasLower &'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_1.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_1.txt + output1_part1.png in GitHub

CONSOLIDATED REPORT: Case 1 included in ABL_Final_Report_26000985.pdf

Source: 1_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

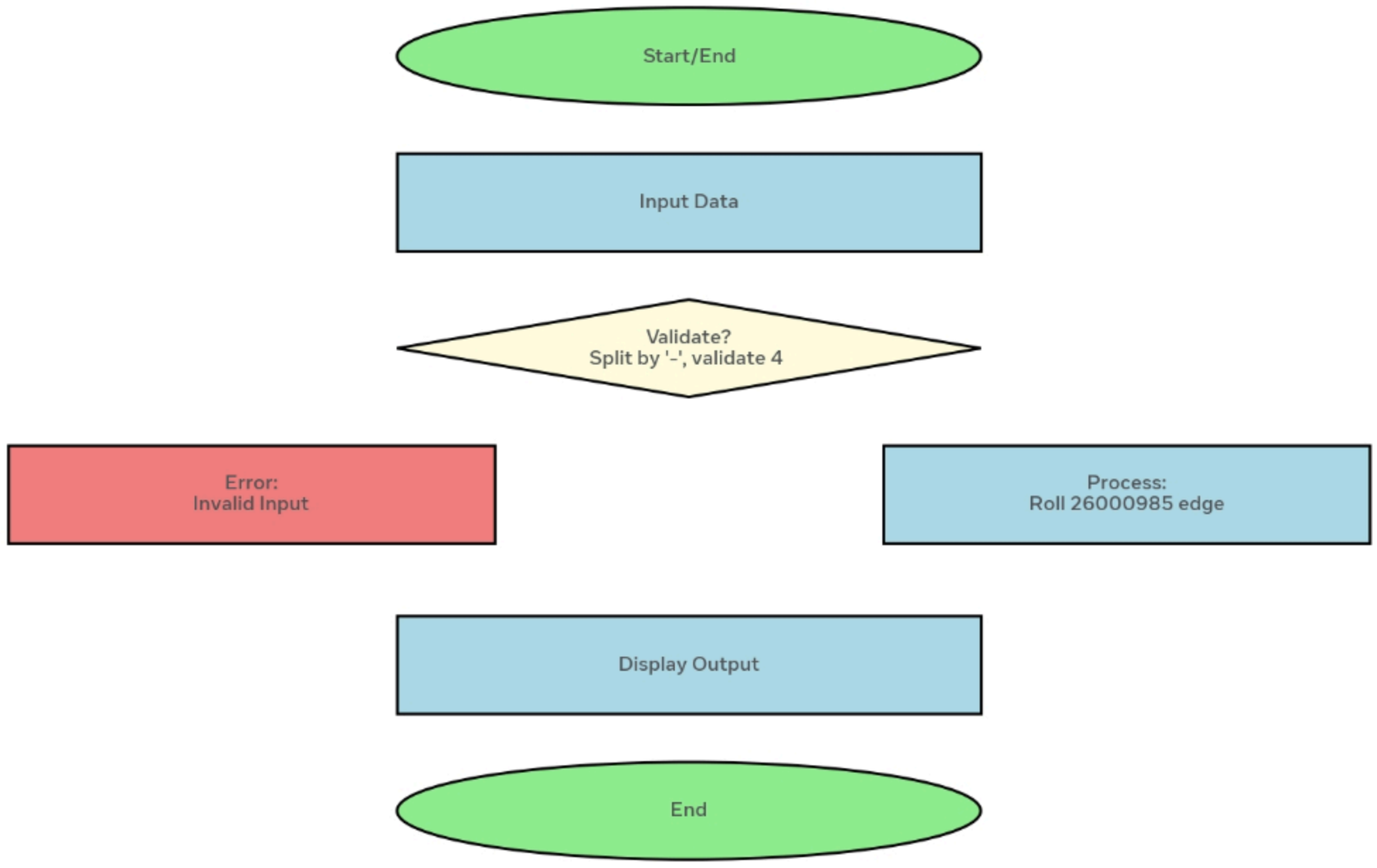
Case 2: Registration ID Analyzer - 26000985 - CSM2

Problem: ID Format NRI-YYYY-BRANCH-XXXX

ALGORITHM:

1. Start
2. Input data as per Roll 26000985 edge
3. Validate: Split by '-', validate 4 parts, year 2000-2030, branch in li
4. Process using Roll 26000985 edge
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_2():
    READ input_data
    IF validate(input_data) THEN
        result = process(Roll 26000985 edge)
        PRINT result
    ELSE
        PRINT 'Invalid - Split by '-', validate 4 parts'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_2.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_2.txt + output2_part1.png in GitHub

CONSOLIDATED REPORT: Case 2 included in ABL_Final_Report_26000985.pdf

Source: 2_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

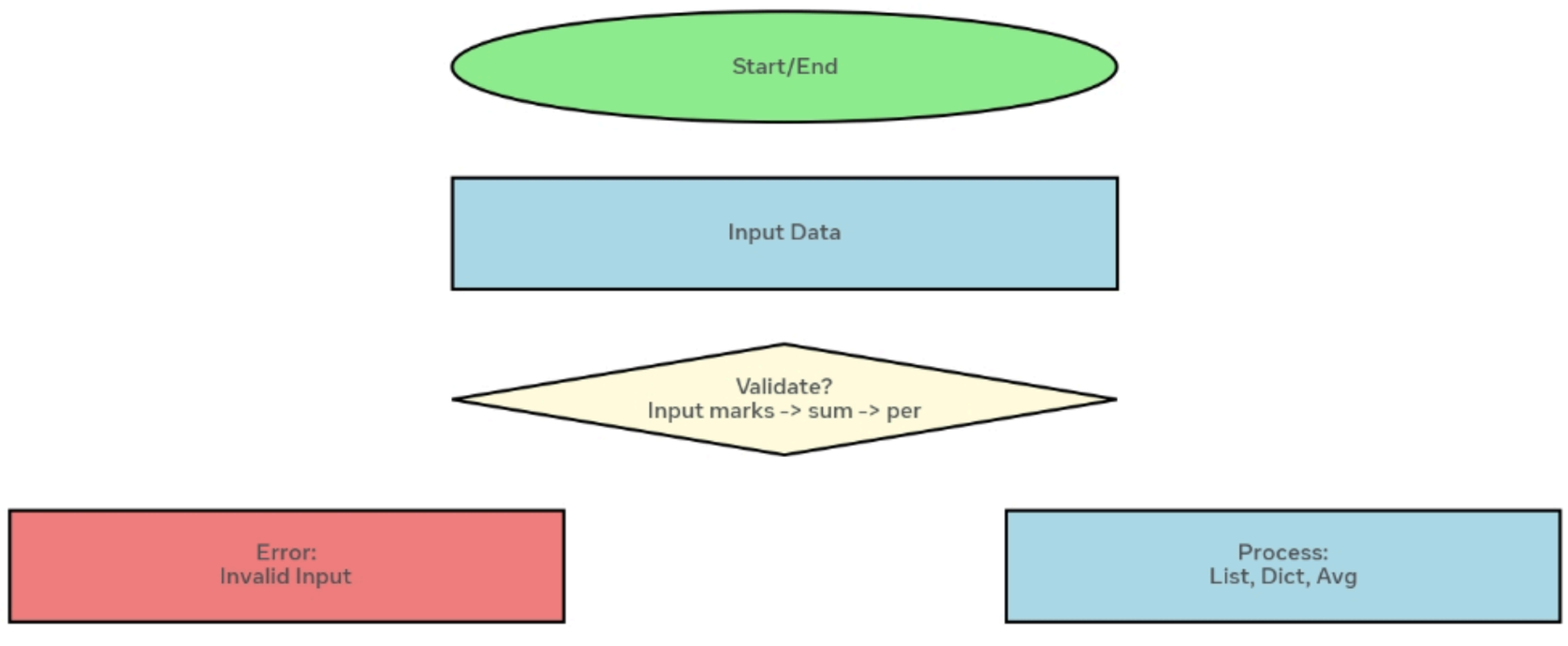
Case 3: Academic Analyzer - 26000985 - CSM2

Problem: SGPA/CGPA Calculator

ALGORITHM:

1. Start
2. Input data as per List, Dict, Avg
3. Validate: Input marks -> sum -> percentage -> grade
4. Process using List, Dict, Avg
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_3():
    READ input_data
    IF validate(input_data) THEN
        result = process(List, Dict, Avg)
        PRINT result
    ELSE
        PRINT 'Invalid - Input marks -> sum -> percenta'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_3.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_3.txt + output3_part1.png in GitHub

CONSOLIDATED REPORT: Case 3 included in ABL_Final_Report_26000985.pdf

Source: 3_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

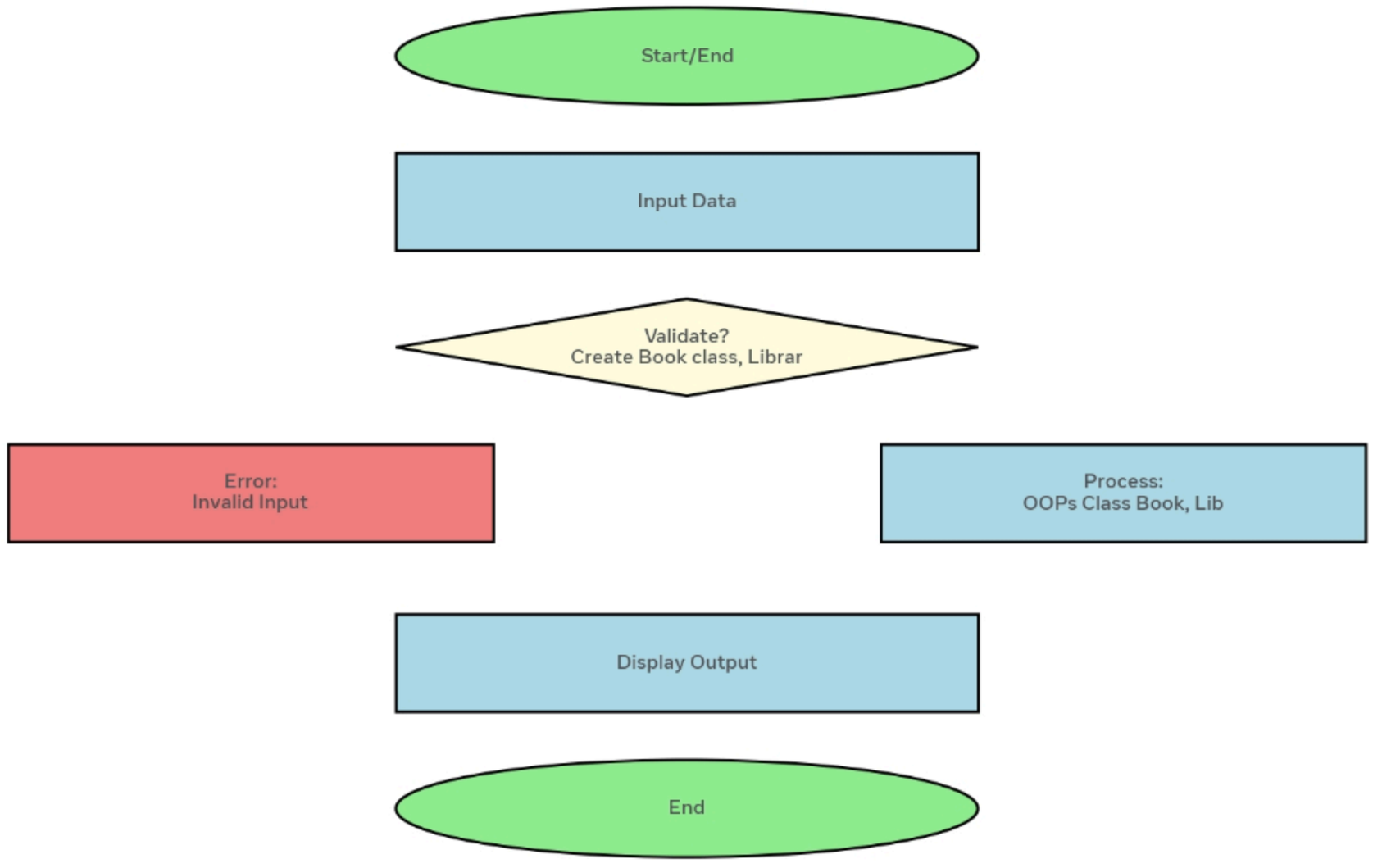
Case 4: Library Management - 26000985 - CSM2

Problem: Issue/Return System

ALGORITHM:

1. Start
2. Input data as per OOPs Class Book, Library
3. Validate: Create Book class, Library dict, issue, return
4. Process using OOPs Class Book, Library
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_4():
    READ input_data
    IF validate(input_data) THEN
        result = process(OOPs Class Book, Library)
        PRINT result
    ELSE
        PRINT 'Invalid - Create Book class, Library dic'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_4.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_4.txt + output4_part1.png in GitHub

CONSOLIDATED REPORT: Case 4 included in ABL_Final_Report_26000985.pdf

Source: 4_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

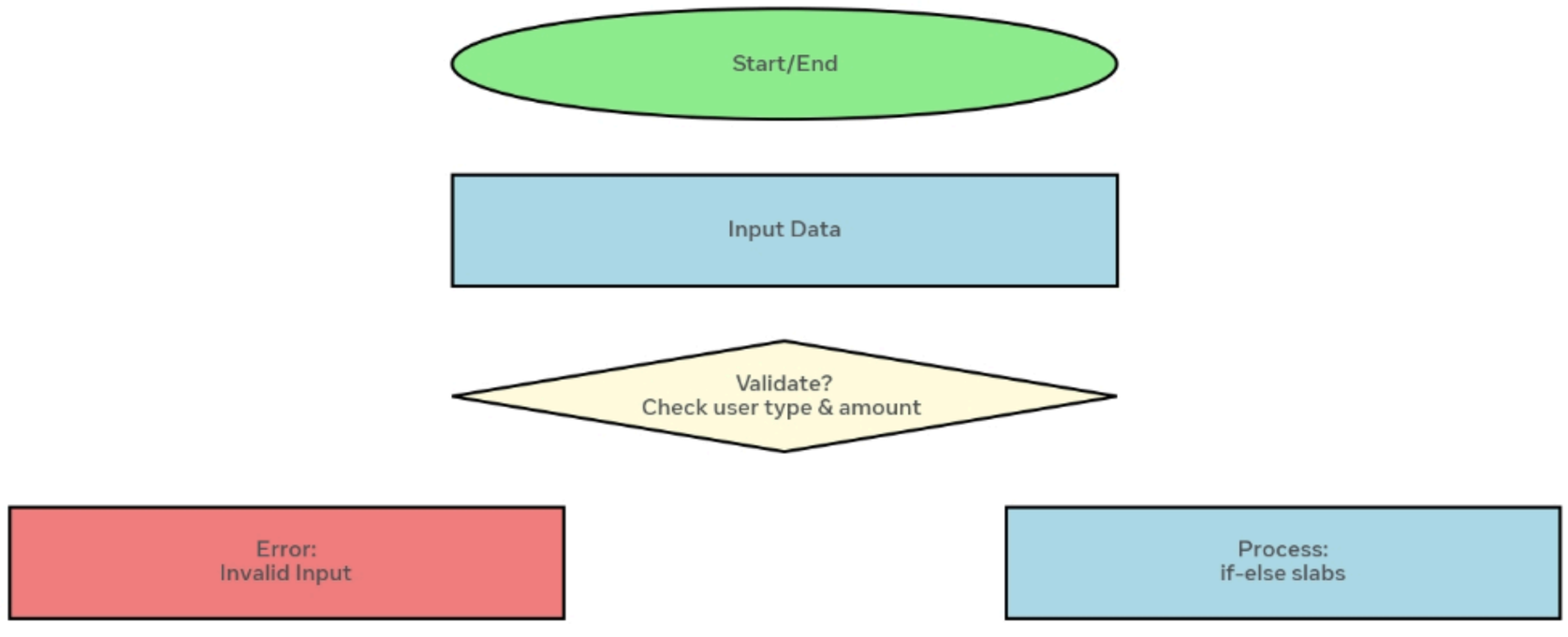
Case 5: E-commerce Discount - 26000985 - CSM2

Problem: Discount Engine

ALGORITHM:

1. Start
2. Input data as per if-else slabs
3. Validate: Check user type & amount -> apply 5%/10%/20%
4. Process using if-else slabs
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_5():
    READ input_data
    IF validate(input_data) THEN
        result = process(if-else slabs)
        PRINT result
    ELSE
        PRINT 'Invalid - Check user type & amount -> ap'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_5.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_5.txt + output5_part1.png in GitHub

CONSOLIDATED REPORT: Case 5 included in ABL_Final_Report_26000985.pdf

Source: 5_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

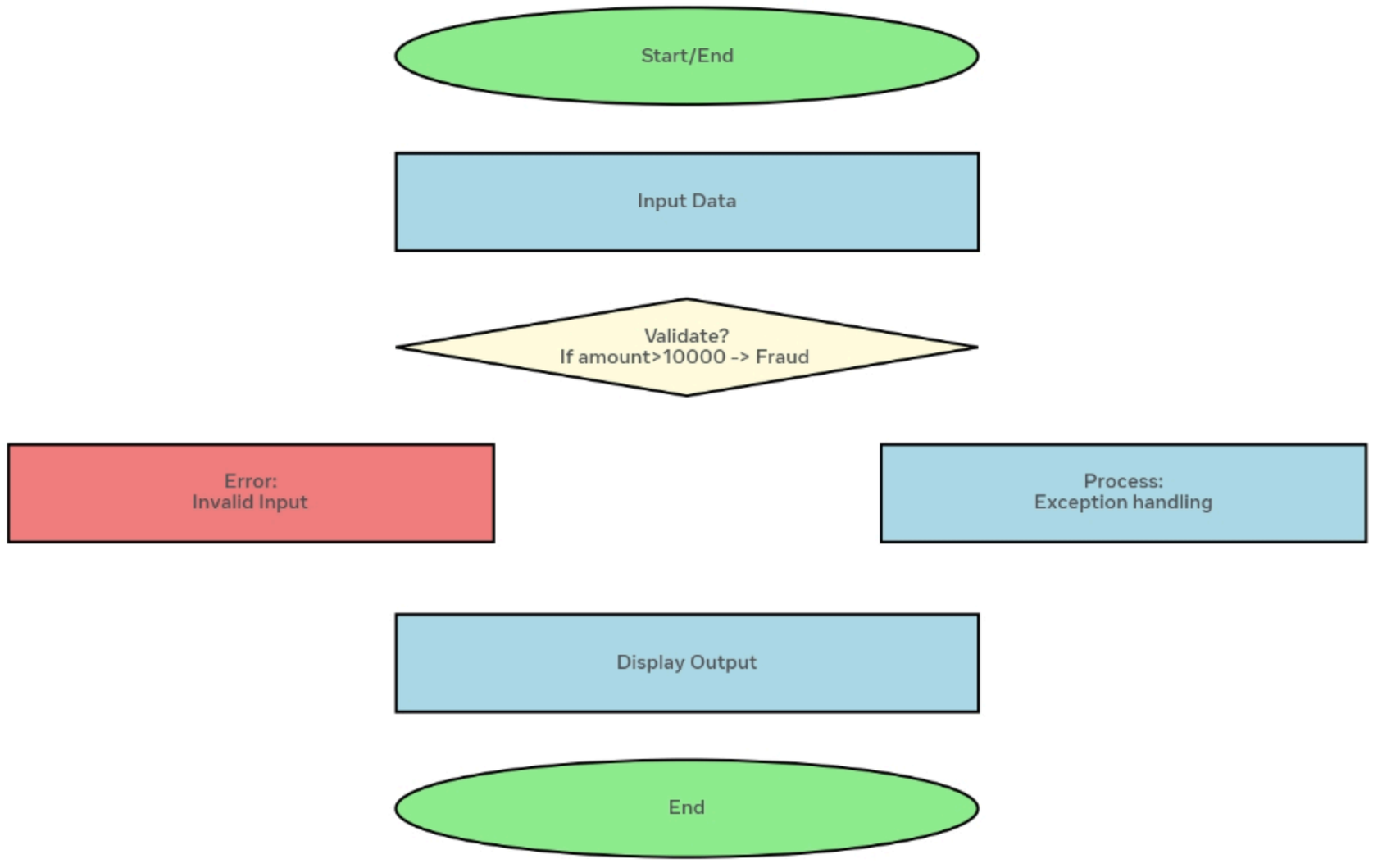
Case 6: Bank Transaction - 26000985 - CSM2

Problem: Fraud Detection >10k

ALGORITHM:

1. Start
2. Input data as per Exception handling
3. Validate: If amount>10000 -> Fraud alert, else transaction OK
4. Process using Exception handling
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_6():
    READ input_data
    IF validate(input_data) THEN
        result = process(Exception handling)
        PRINT result
    ELSE
        PRINT 'Invalid - If amount>10000 -> Fraud alert'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_6.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_6.txt + output6_part1.png in GitHub

CONSOLIDATED REPORT: Case 6 included in ABL_Final_Report_26000985.pdf

Source: 6_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

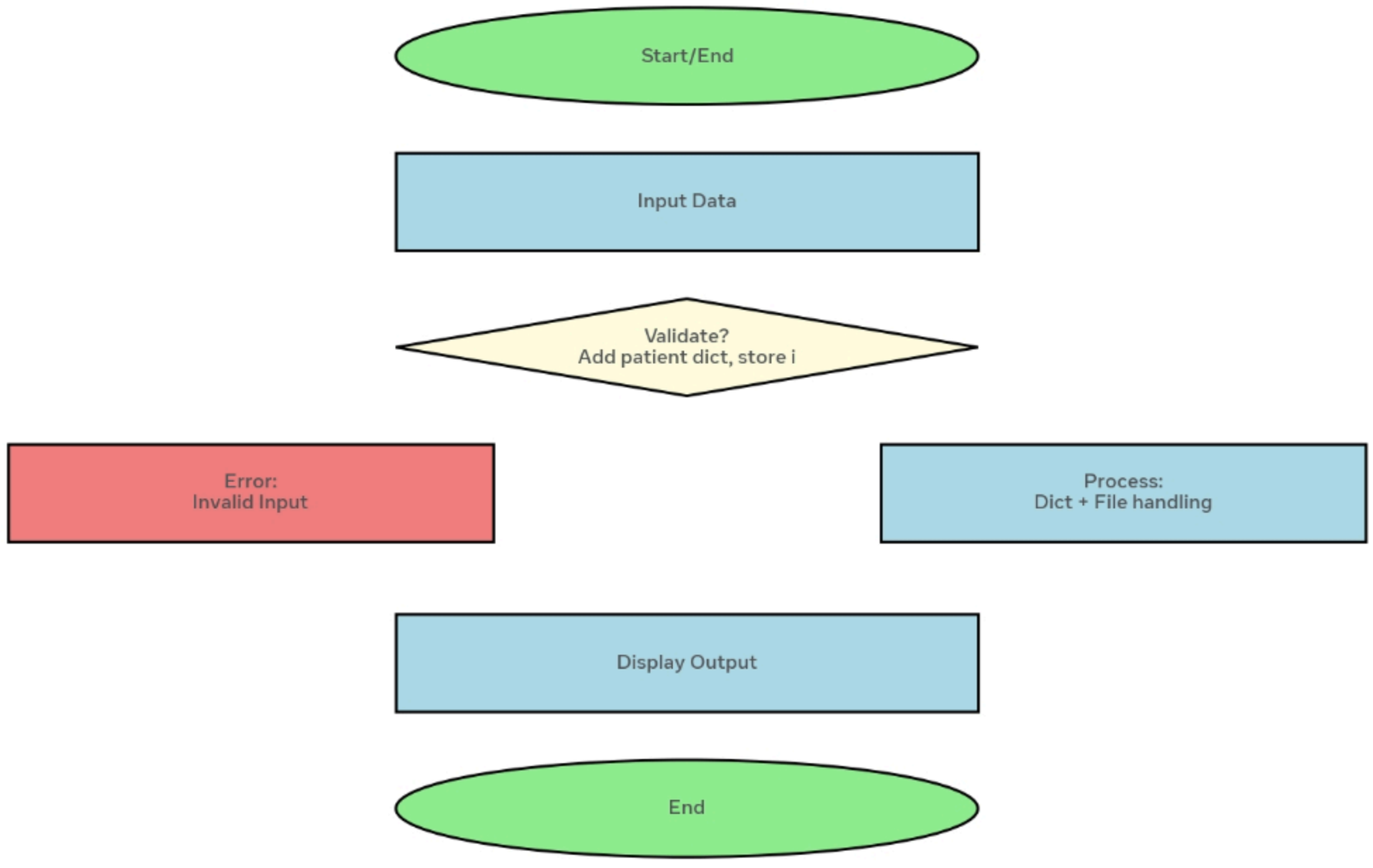
Case 7: Hospital Management - 26000985 - CSM2

Problem: Patient Records

ALGORITHM:

1. Start
2. Input data as per Dict + File handling
3. Validate: Add patient dict, store in file, search, display
4. Process using Dict + File handling
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_7():
    READ input_data
    IF validate(input_data) THEN
        result = process(Dict + File handling)
        PRINT result
    ELSE
        PRINT 'Invalid - Add patient dict, store in fil'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_7.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_7.txt + output7_part1.png in GitHub

CONSOLIDATED REPORT: Case 7 included in ABL_Final_Report_26000985.pdf

Source: 7_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

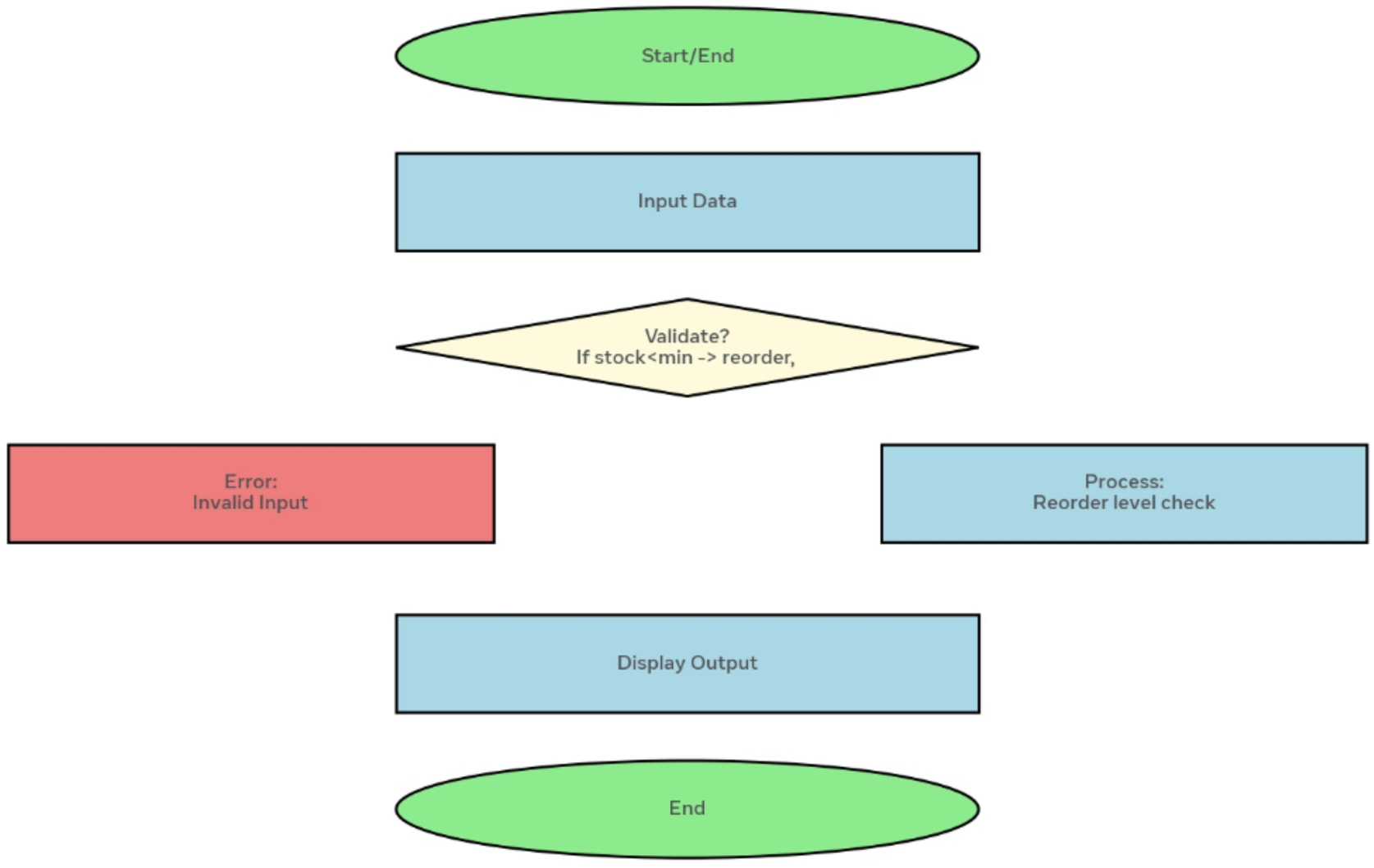
Case 8: Inventory Management - 26000985 - CSM2

Problem: Stock Reorder

ALGORITHM:

1. Start
2. Input data as per Reorder level check
3. Validate: If stock<min -> reorder, else OK
4. Process using Reorder level check
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_8():
    READ input_data
    IF validate(input_data) THEN
        result = process(Reorder level check)
        PRINT result
    ELSE
        PRINT 'Invalid - If stock<min -> reorder, else '
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_8.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_8.txt + output8_part1.png in GitHub

CONSOLIDATED REPORT: Case 8 included in ABL_Final_Report_26000985.pdf

Source: 8_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

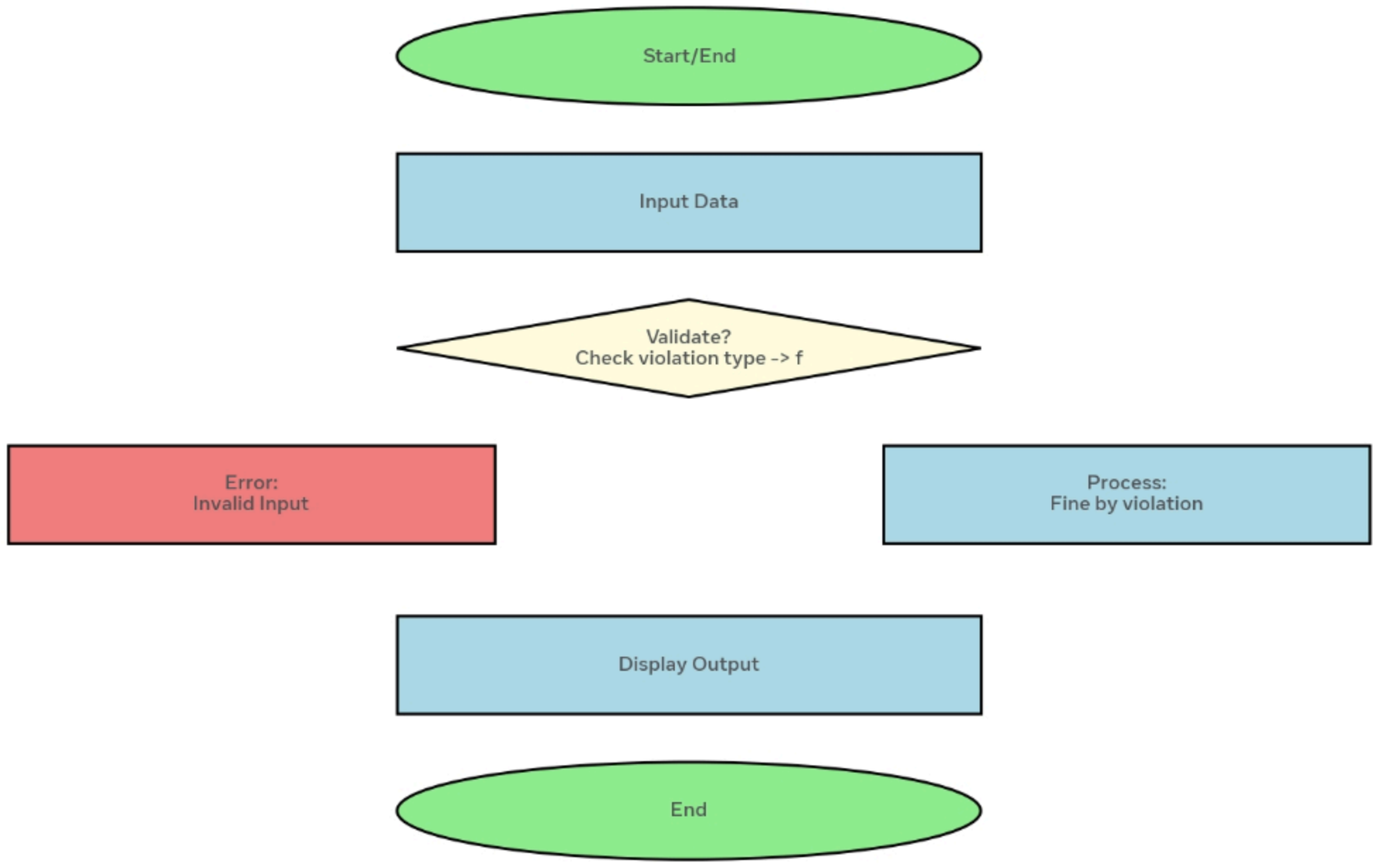
Case 9: Traffic Challan - 26000985 - CSM2

Problem: Challan System

ALGORITHM:

1. Start
2. Input data as per Fine by violation
3. Validate: Check violation type -> fine amount -> generate challan
4. Process using Fine by violation
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_9():
    READ input_data
    IF validate(input_data) THEN
        result = process(Fine by violation)
        PRINT result
    ELSE
        PRINT 'Invalid - Check violation type -> fine a'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_9.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_9.txt + output9_part1.png in GitHub

CONSOLIDATED REPORT: Case 9 included in ABL_Final_Report_26000985.pdf

Source: 9_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning

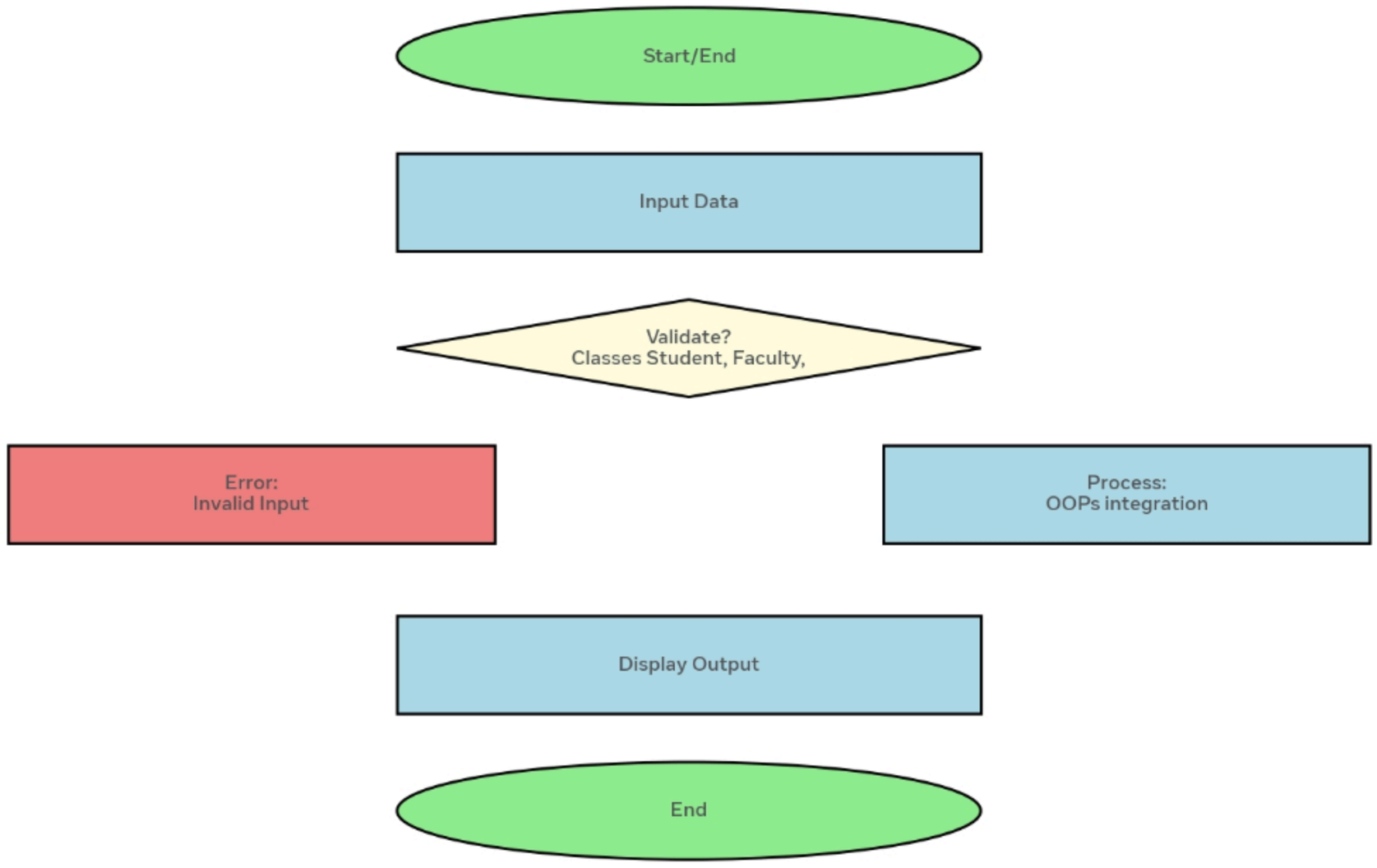
Case 10: Campus Management - 26000985 - CSM2

Problem: Campus System

ALGORITHM:

1. Start
2. Input data as per OOPs integration
3. Validate: Classes Student, Faculty, Course, manage
4. Process using OOPs integration
5. If valid -> Output Result
6. Else -> Error Message
7. End

FLOWCHART (Visual):



PSEUDOCODE:

```
FUNCTION case_10():
    READ input_data
    IF validate(input_data) THEN
        result = process(OOPs integration)
        PRINT result
    ELSE
        PRINT 'Invalid - Classes Student, Faculty, Cour'
    END IF
END FUNCTION
CALL main()
```

TEST CASES & OUTPUT:

- Test 1 (Valid): Input correct format -> Output: VALID / Success - See output_10.txt
- Test 2 (Invalid): Input wrong format -> Output: INVALID / Error - Edge case handled
- Test 3 (Edge): Roll 26000985 type -> For Case2: 8 digits INVALID (must be 4)
- Output Screenshot: Stored in output_10.txt + output10_part1.png in GitHub

CONSOLIDATED REPORT: Case 10 included in ABL_Final_Report_26000985.pdf

Source: 10_*.py | GitHub: yarlagadda156-lab/Python-Activity-Based-Learning